# SISTEM BILANGAN

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# MATERI

PENGERTIAN SISTEM BILANGAN

JENIS – JENIS SISTEM BILANGAN

- DESIMAL
- BINER
- OKTAL
- HEKSADESIMAL

KONVERSI JENIS-JENIS SISTEM BILANGAN

# SISTEM BILANGAN

SISTEM BILANGAN (NUMBER SYSTEM) ADALAH SUATU CARA UNTUK MEWAKILI BESARAN DARI SUATU “ITEM” FISIK.

# JENIS SISTEM BILANGAN

POLBAN

## DESIMAL (Basis 10)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

**180** (10)
**257** (DES)

## BINER (Basis 2)

| 0 | 1 |
|---|---|

**1011** (2)
**1011** (BIN)

## OKTAL (Basis 8)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|

**150** (8)
**257** (OKT)

## HEKSADESIMAL (Basis 16)

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**5C** (16)
**1A** (HEX)

# SISTEM BILANGAN DESIMAL

## BASIS 10

D = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }

Setiap angka disebut dengan DIGIT

Contoh :

$2\ 4\ 5_{(10)} \rightarrow (2 \times 10^2) + (4 \times 10^1) + (5 \times 10^0)$

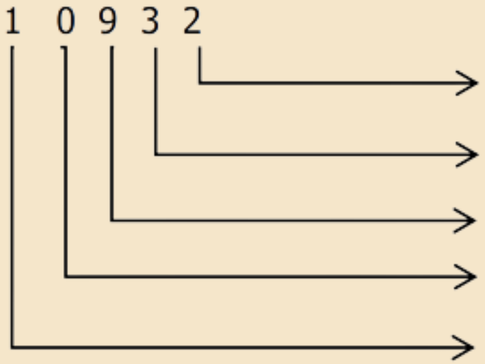
MSD

LSD

MSD : Most significant Digit

LSD : Least significant Digit

Angka desimal 10932 ($10932_{(10)}$)




| | | | |
|---|---|---|---|
| Pertama | $2 . 10^0 = 2 . 1$ | = | 2 |
| Kedua | $3 . 10^1 = 3 . 10$ | = | 30 |
| Ketiga | $9 . 10^2 = 9 . 100$ | = | 900 |
| Keempat | $0 . 10^3 = 0 . 1000$ | = | 0 |
| Kelima | $1 . 10^4 = 1 . 10000$ | = | 10000 |
| | | | 10932 |

Contoh :

$2\ 4\ 5\ ,\ 2\ 3_{(10)} = (2 \times 10^2) + (4 \times 10^1) + (5 \times 10^0) + (2 \times 10^{-1}) + (3 \times 10^{-2})$

$= 200 + 40 + 5 + 0{,}2 + 0{,}03$

MSD

LSD

# SISTEM BILANGAN BINER

## BASIS 2 ( 0, 1 )

**BINARY DIGIT (BIT)**

4 BIT = 1 NIBBLE

8 BIT = 1 BYTE

16 BIT = 1 WORD

**CONTOH**

1011 (2)




**MSB : Most Significant Bit**

**LSB : Least Significant Bit**

**KONVERSI BINER - DESIMAL**

$1011_{(2)} = ( 1 \times 2^3 ) + ( 0 \times 2^2 ) + ( 1 \times 2^1 ) + ( 1 \times 2^0 )$

$= 8 + 0 + 2 + 1$

$= 11_{(10)}$

$11{,}01_{(2)} = ( 1 \times 2^1 ) + ( 1 \times 2^0 ) + ( 0 \times 2^{-1} ) + ( 1 \times 2^{-2} )$

$= 2 + 1 + 0 + 0{,}25$

$= 3, 25_{(10)}$

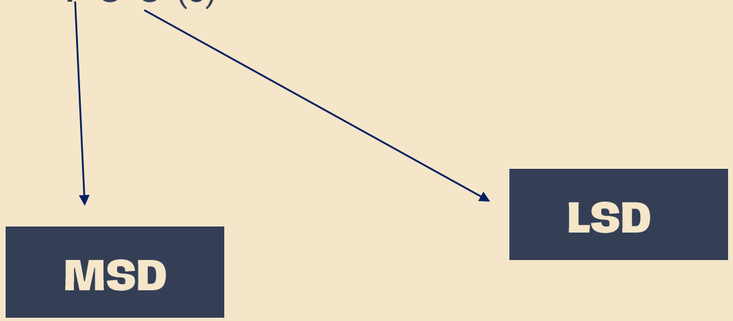
# SISTEM BILANGAN OKTAL

## BASIS 8

D = {0, 1, 2, 3, 4, 5, 6, 7}

Contoh :

1 5 0 (8)




MSD : Most significant Digit
LSD : Least significant Digit

## KONVERSI OKTAL - DESIMAL

$1\ 5\ 0_{(8)} = (1 \times 8^2) + (5 \times 8^1) + (0 \times 8^0)$

$= 64 + 40 + 0$

$= 104_{(10)}$

$3\ 4\ 5,4\ 4_{(8)} = (3 \times 8^2) + (4 \times 8^1) + (5 \times 8^0) + (4 \times 8^{-1}) + (4 \times 8^{-2})$

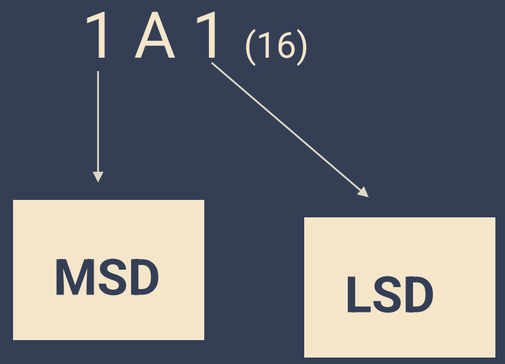
$= 192 + 32 + 5 + 0.5 + 0.0625$

$= 2\ 2\ 9,5\ 6\ 2\ 5_{(10)}$

# SISTEM BILANGAN HEKSADESIMAL

## BASIS 16

{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F }




MSB : Most Significant Digit

LSB : Least Significant Digit

| HEKSADESIMAL | DESIMAL |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |
| 4 | 4 |
| 5 | 5 |
| 6 | 6 |
| 7 | 7 |
| 8 | 8 |
| 9 | 9 |
| A | 10 |
| B | 11 |
| C | 12 |
| D | 13 |
| E | 14 |
| F | 15 |

## KONVERSI HEKSADESIMAL - DESIMAL

$1\ A\ 1_{(16)} = (1 \times 16^2) + (10 \times 16^1) + (1 \times 16^0)$

$= 256 + 160 + 1$

$= 417_{(10)}$

$2\ A\ ,\ C_{(16)} = (2 \times 16^1) + (10 \times 16^0) + (12 \times 16^{-1})$

$= 32 + 10 + 0{,}75$

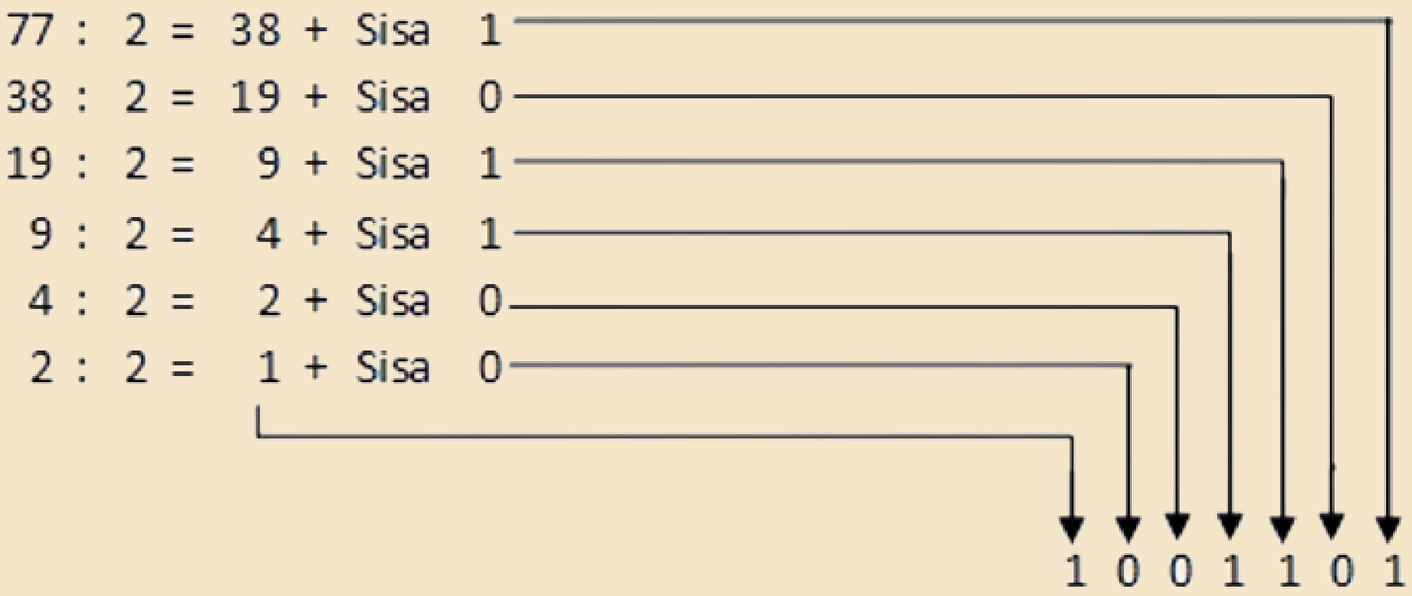
$= 4\ 2\ ,\ 7\ 5_{(10)}$

# KONVERSI DESIMAL KE BINER

**Cara : Membagi bilangan desimal dengan nilai basis Biner yaitu 2.**

**Contoh :**

**77(10) = …….(2)**




$98_{10} = \ldots\ldots_{2}$

- 98 : 2 = 49 sisa **0**
- 49 : 2 = 24 sisa **1**
- 24 : 2 = 12 sisa **0**
- 12 : 2 = 6 sisa **0**
- 6 : 2 = 3 sisa **0**
- 3 : 2 = **1** sisa **1**

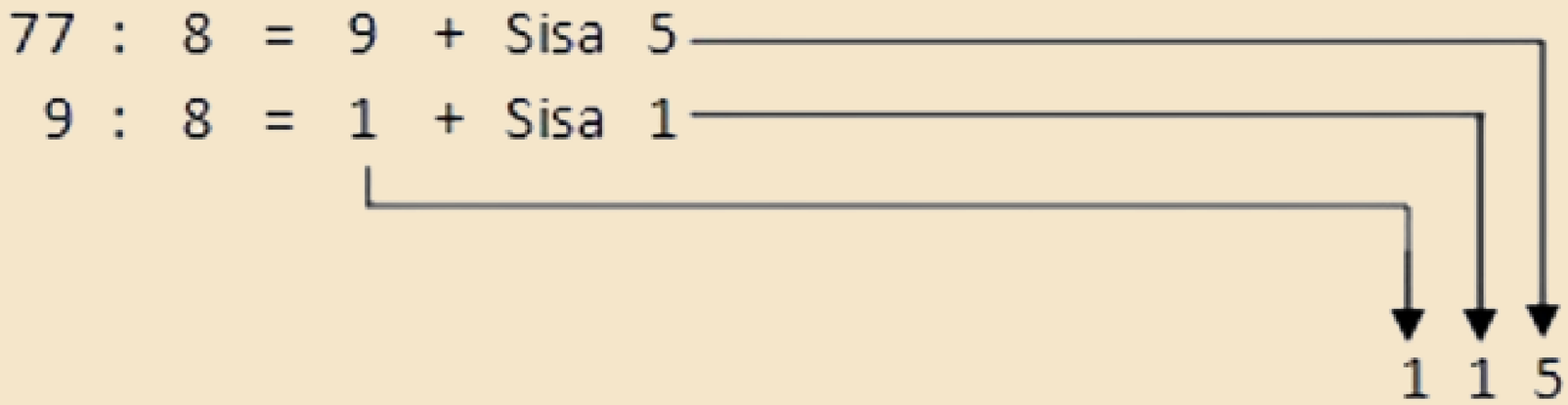
$1100010_2$

# KONVERSI DESIMAL KE OKTAL

**Cara : Membagi bilangan desimal dengan nilai basis Oktal yaitu 8**

**Contoh :**

**$77_{(10)}$ = .....(8)**




→ **$77_{(10)} = 115_{(8)}$**

# KONVERSI DESIMAL KE HEKSADESIMAL

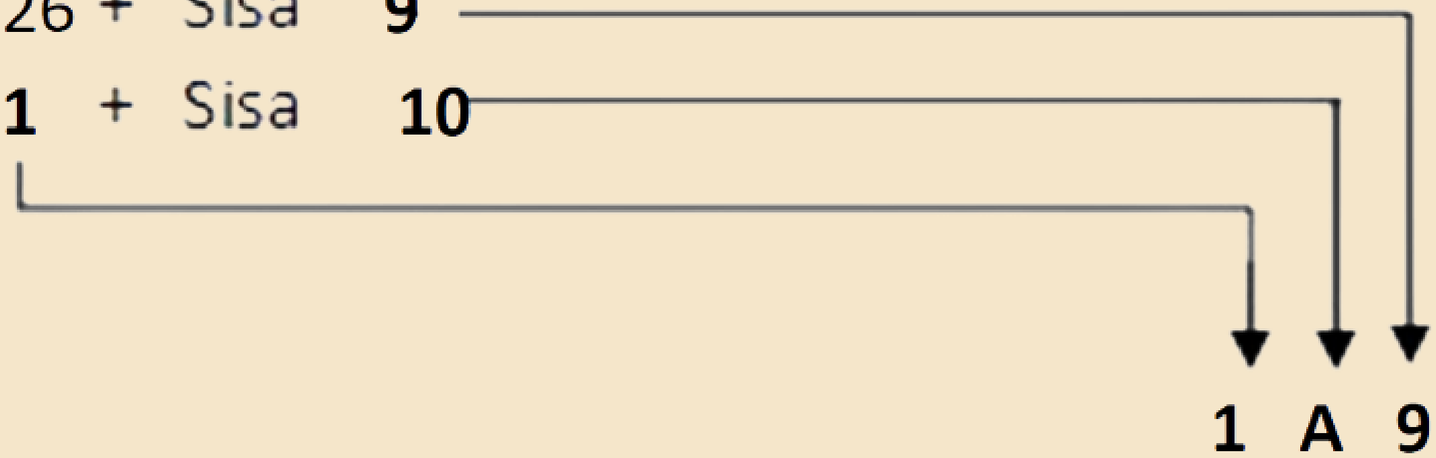
**Cara : Membagi bilangan desimal dengan nilai basis Heksadesimal yaitu 16**

**Contoh**

**$425_{(10)}$ = ……$_{(16)}$**

425 : 16 = 26 + Sisa **9**

26 : 16 = **1** + Sisa **10**



**1 A 9**

→ **$425_{(10)} = 1A9_{(16)}$**

# KONVERSI DARI BILANGAN BINER

**1-bit**

$0_{10} = 0_2$
$1_{10} = 1_2$

**2-Keadaan**

**2-bit**

$0_{10} = 00_2$
$1_{10} = 01_2$
$2_{10} = 10_2$
$3_{10} = 11_2$

**4-Keadaan**

**3-bit**

$0_{10} = 000_2$
$1_{10} = 001_2$
$2_{10} = 010_2$
$3_{10} = 011_2$
$4_{10} = 100_2$
$5_{10} = 101_2$
$6_{10} = 110_2$
$7_{10} = 111_2$

**8-Keadaan**

**BASIS 8**
**BIL. OKTAL**

**4-bit**

$0_{10} = 0000_2$
$1_{10} = 0001_2$
$2_{10} = 0010_2$
$3_{10} = 0011_2$
$4_{10} = 0100_2$

...

$15_{10} = 1111_2$

**16-Keadaan**

**BASIS 16**
**BIL. HEKSADESIMAL**

# KONVERSI BINER KE OKTAL DAN HEKSADESIMAL

## BINER - OKTAL

$11110101_{(2)} = \ldots\ldots.{}_{(8)}$

$\underset{3}{\underline{011}}\ \underset{6}{\underline{110}}\ \underset{5}{\underline{101}}_{(2)} = 365_{(8)}$

## BINER - HEKSADESIMAL

$11110101_{(2)} = \ldots\ldots.{}_{(8)}$

$\underset{15}{\underline{1111}}\ \underset{5}{\underline{0101}}_{(2)} = F5_{(8)}$

# KONVERSI OKTAL KE BINER DAN HEKSADESIMAL

## OKTAL - BINER

4573(8) = .....(2)

4 → 100

5 → 101

7 → 111

3 → 011

**4573(8) = 100101111011(2)**

## OKTAL - HEKSADESIMAL

756(8) = .....(16)

7 → 111

5 → 101

6 → 110

111101110 (2)

**0001 1110 1110 = 1EE (16)**

**1 14 14**

# KONVERSI HEKSADESIMAL KE BINER DAN OKTAL

## HEKSADESIMAL - BINER

**A51** (16) **=** **.....**(2)

A = 10 → 1010

5 → 0101

1 → 0001

**A51** (16) **= 101001010001**(2)

## HEKSADESIMAL - OKTAL

**1A5**(16) = .....(8)

1 → 0001

A → 1010

5 → 0101

110100101 (2)

**110 100 101 = 645** (8)

**6 4 5**